Table 1. Data sources

Dataset / endpoint	Cases	Controls	Exposure or clinical role
eQTLGen blood expression	—	—	Exposure: up to 31,684; predominantly European
BBJ GD (GCST90018627)	2,809	172,656	Discovery; East Asian
FinnGen GD (E4_GRAVES_STRICT)	3,962	496,386	Post hoc GD comparison; Finnish
UKB hyperthyroidism (GCST90038636)	3,731	480,867	Self-report; ancestry not established as exclusively European
FinnGen GO (GRAVES_OPHT)	858	499,490	Original ophthalmopathy comparison; Finnish
FinnGen GO strict (E4_GRAVES_OPHT_STRICT)	753	499,595	Post hoc definition sensitivity; Finnish

GD, Graves disease; GO, Graves ophthalmopathy; HT, hyperthyroidism/thyrotoxicosis. The three FinnGen R12 endpoints overlap and use population non-cases for their respective definitions. Strict is the source endpoint label, not proof of perfect specificity. The descriptive orbital series (four TED patients, one control) is separate from the genetic datasets.

Table 2. TSHR, IGF1R and CTLA4 across five disease datasets

Gene	Dataset	IVs	OR (95% CI)	P value	PP.H3	PP.H4
TSHR	BBJ GD	1	0.12 (0.07–0.21)	**1.09×10^{-14}**	0.049	**0.951**
TSHR	FinnGen GD	1	0.08 (0.05–0.12)	**3.19×10^{-32}**	0.736	0.264
TSHR	UKB HT	1	0.09 (0.06–0.14)	**8.77×10^{-28}**	0.774	0.226
TSHR	FinnGen GO	1	0.10 (0.04–0.24)	**2.82×10^{-7}**	0.014	**0.986**
TSHR	FinnGen GO strict	1	0.15 (0.06–0.39)	**9.02×10^{-5}**	0.082	**0.891**
IGF1R	BBJ GD	4	1.56 (1.07–2.28)	**0.0212**	0.236	0.073
IGF1R	FinnGen GD	3	1.07 (0.86–1.33)	0.539	0.259	0.010
IGF1R	UKB HT	4	1.35 (1.07–1.70)	**0.0117**	0.196	0.404
IGF1R	FinnGen GO	3	1.41 (0.85–2.33)	0.182	0.346	0.032
IGF1R	FinnGen GO strict	3	1.52 (0.92–2.49)	0.0996	0.391	0.038
CTLA4	BBJ GD	1	0.18 (0.11–0.27)	**5.45×10^{-15}**	0.799	0.201
CTLA4	FinnGen GD	2	0.17 (0.08–0.36)	**2.15×10^{-6}**	0.746	0.254
CTLA4	UKB HT	2	0.21 (0.08–0.57)	**0.0022**	0.047	**0.953**
CTLA4	FinnGen GO	2	0.17 (0.04–0.68)	**0.0118**	0.022	**0.978**
CTLA4	FinnGen GO strict	2	0.17 (0.05–0.64)	**0.00888**	0.021	**0.979**

IVs, instrumental variables (retained genetic variants). Wald ratio for one IV; random-effects IVW for multiple IVs. ORs are per approximately one SD higher reconstructed blood expression. Bold P values denote nominal $P < 0.05$ only. Discovery uses 0.05/2,544. PP.H3 supports distinct variants; PP.H4 supports one shared variant. Bold PP.H4 denotes ≥0.80 under $p12 = 10^{-5}$. These are probabilities, not P values. CTLA4 multi-IV results are concentrated in rs13030124; TSHR is single-IV in all datasets.

Table 3. Screen and supporting analyses — A

BBJ discovery gene	BBJ OR (95% CI)	BBJ P	FinnGen GD OR (95% CI)	FinnGen GD P	Focused support	GD PP.H4
HLA-A	1.98 (1.73–2.27)	**2.60×10^{-23}**	1.08 (0.77–1.53)	0.648	No	<0.001
HLA-DQA2	2.39 (1.95–2.93)	**8.30×10^{-17}**	1.90 (1.64–2.20)	**1.41×10^{-17}**	Yes	<0.001
CTLA4	0.18 (0.11–0.27)	**5.45×10^{-15}**	0.17 (0.08–0.36)	**2.15×10^{-6}**	Yes	0.254
TSHR	0.12 (0.07–0.21)	**1.09×10^{-14}**	0.08 (0.05–0.12)	**3.19×10^{-32}**	Yes	0.264
C4A	0.45 (0.35–0.57)	**2.23×10^{-10}**	0.41 (0.26–0.63)	**4.97×10^{-5}**	Yes	<0.001
HSD3B7	0.29 (0.18–0.47)	**2.04×10^{-7}**	0.90 (0.72–1.13)	0.372	No	0.011
TUBB	1.49 (1.28–1.74)	**3.14×10^{-7}**	3.28 (2.09–5.17)	**2.68×10^{-7}**	Yes	<0.001
VKORC1	0.13 (0.06–0.29)	**6.35×10^{-7}**	NA	NA	NA	0.011
TNFSF14	0.63 (0.52–0.76)	**1.48×10^{-6}**	0.77 (0.65–0.92)	**0.00361**	Yes	0.303
PRSS36	6.00 (2.87–12.57)	**2.02×10^{-6}**	1.40 (1.00–1.96)	**0.049**	No	0.063
MAPKAPK5	0.977 (0.967–0.987)	**5.28×10^{-6}**	0.990 (0.929–1.055)	0.751	No	0.041
PSMB8	1.25 (1.13–1.38)	**6.86×10^{-6}**	0.96 (0.85–1.08)	0.512	No	<0.001
IFNGR1	2.10 (1.51–2.91)	**9.41×10^{-6}**	1.62 (1.18–2.24)	**0.0033**	Yes	0.313

Panel A: focused support requires concordant direction and $P < 0.05/13$; 7 of 13 qualify. NA is unavailable, not a negative result. Bold P values denote nominal $P < 0.05$; screen and focused thresholds are stated separately.

Table 3. Screen and supporting analyses — B

Additional gene	GD IVs	GD OR (95% CI)	GD P	BBJ PP.H4	GD PP.H4	GO strict PP.H4
APOM	1	0.05 (0.04–0.06)	**1.43×10^{-108}**	<0.001	<0.001	0.227
NEU1	1	0.04 (0.03–0.06)	**1.43×10^{-108}**	<0.001	<0.001	0.061
IGLV2-18	1	0.56 (0.48–0.66)	**7.52×10^{-12}**	0.029	0.020	0.100
IGLV3-12	1	1.20 (1.14–1.26)	**7.52×10^{-12}**	0.019	0.022	0.101
TNF	1	0.60 (0.51–0.71)	**4.65×10^{-10}**	<0.001	<0.001	<0.001
HIST1H4E	1	2.54 (1.88–3.43)	**1.51×10^{-9}**	<0.001	<0.001	0.007
LTA	1	0.46 (0.35–0.61)	**3.75×10^{-8}**	<0.001	<0.001	<0.001
LTB	1	0.18 (0.10–0.36)	**4.89×10^{-7}**	<0.001	<0.001	<0.001
C4B	2	1.63 (1.31–2.03)	**1.10×10^{-5}**	<0.001	<0.001	0.311

nine additional reference-frequency FinnGen GD discoveries at $P < 0.05/2{,}544$; post hoc and not independent of the full screen. APOM/NEU1 share rs3101018; IGLV2-18/IGLV3-12 share rs5759410. The lead IGLV expression variant is absent in BBJ, weakening interpretation of its low PP.H4.

Table 3. Screen and supporting analyses — C

Gene / GD analysis	Reference OR (95% CI)	Reference P	Cohort-frequency OR (95% CI)	Cohort P	Interpretation
C4B	1.63 (1.31–2.03)	**1.10×10^{-5}**	1.75 (1.31–2.33)	**1.48×10^{-4}**	Heterogeneous; loses screen significance
HLA-G	0.86 (0.72–1.03)	0.107	0.60 (0.47–0.75)	**1.72×10^{-5}**	Weight-sensitive; single-variant dependence

HLA-G was added because of frequency sensitivity, not primary discovery. Neither C4B nor HLA-G gains strong GD colocalization. All 24 genes and five datasets are available in the workbook. Bold P values denote nominal $P < 0.05$; screen and focused thresholds are stated separately.